



**ADDITION OF BM LINE 3 TO THE EXISTING AUTO-X OPERATIONS, NEAVE
TOWNSHIP (KORSTEN) IN GQEBERHA WITHIN THE NELSON MANDELA
BAY MUNICIPALITY, EASTERN CAPE PROVINCE**

ENVIRONMENTAL MANAGEMENT PROGRAMME

DEDEAT REFERENCE NO: ECm1/C/LN1/M/36-2026

24 August 2026

Applicant:



Environmental Assessment Practitioner:

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DOCUMENT REVISION STATUS

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	19 August 2026	24 August 2026	24 August 2026

DECLARATION OF INDEPENDENCE

I, Zikhona I. Wana, in my capacity as an EAP, hereby declare that I:

- Act as an independent consultant;
- Do not have any financial interest in the undertaking of the activity, other than remuneration for the work performed;
- Have no and will not have vested interest in the activity;
- Have no and will not engage in conflicting interests in the undertaking of the activity;
- Undertake to disclose any material information that has or may have the potential to influence the decision of the CA or the objectivity of any report, plan, or document;
- Will provide the CA with access to all information at my disposal regarding the report, whether such information is favourable to the Client or not;
- Based on information provided to me by the client, and in addition to information obtained during the course of this study, have presented the results and conclusion within the associated document to the best of my professional ability;
- Reserve the right to modify aspects pertaining to the present investigation should additional information become available through on-going research and/ or further work in this field; and
- Undertake to have my work peer reviewed on a regular basis by a competent specialist.



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DETAILS AND EXPERTISE OF THE EAP

Expertise:	Zikhona is a <i>Reg. EAP</i> with the EAPASA, Registration No. 2019/555, and registered as a <i>Pr.Sci.Nat</i> with the SACNASP in the field of environmental science, Registration No. 119417. The service provider has over 12 years of environmental consulting experience. She has undertaken the EA application process in both the BA and Scoping and Environmental Impact Assessment way in several projects. She has also gained experience in other environmental management services undertaking a few responsibilities that include Screening/ Prefeasibility Assessments, Impact Assessments and Drafting of EMPrs as per the NEMA. She has also gained experience in undertaking of Waste Storage Registrations, Waste Handling (sorting, screening, shredding, etc.) Registrations, Waste Classification/ Characterisation, Waste Management Licence Applications, and Waste Site Decommissioning processes as is relevant under the National Environmental Management Waste Act 59 of 2008 as amended (NEMWA). Zikhona has executed several Public Participation Processes as per the EIA Regulations, NEMWA, and the National Water Act 36 of 1998, as amended (NWA). Her other assignments included Integrated Water and Waste Management Plans. Her experience also includes undertaking General Authorisations, Water Use Licence Applications, Integrated Water Use Licence Applications and production of associated reports in terms of NWA. Her roles have also exposed her to compiling terms of reference for specialists, commissioning of specialists, and their management as well as the review of specialist reports for alignment with the scope of work and legislative requirements. Her other capabilities include Phase 1 Environmental Due Diligence, Environmental Management Systems, Project Sustainability Management systems, Waste Management Plans, Emergency Preparedness Plans. Qualifications • BTech Environmental Sciences • NDip Environmental Sciences
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UNDERTAKING (AFFIRMATION BY THE EAP)

Abantu hereby confirms that, to the best of our knowledge, the information provided in this report was correct at the time of compilation. Information included in this report was based on the information which was provided to us by the Applicant and the commissioned specialist. It is further confirmed that all comments received from stakeholders and interested and affected parties will be included in the final BAR to be submitted to the CA for review and decision-making.

ASSUMPTIONS AND GAPS IN KNOWLEDGE

The following assumptions and potential gaps in knowledge apply:

- All information provided by the applicant to the EAP was correct and valid at the time it was provided.
- The EAP does not accept any responsibility in the event that additional information comes to light at a later stage after the process.
- All data from unpublished research is valid and accurate at the time this report was written.
- The scope of this investigation is limited to assessing the potential environmental impacts associated with the proposed addition of a third battery machine line (BM Line 3) for the assembly of lead-acid batteries.

It should be noted that findings, recommendations, and conclusions provided in this report are based on the author's best scientific and professional knowledge, and experience. No part of this report may be amended or extended without prior written consent of the author. Any recommendations, statements or conclusions drawn from or based on this report must clearly cite or refer to this report. Whenever such recommendations, statements, or conclusions form part of the main report to current investigation, this report must be included in its entirety.

ACKNOWLEDGEMENTS

The authors acknowledge Auto-X for their assistance with project information, facility illustration, and the associated project background information documents as well as for responding to technical queries related to the project.

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TERMS, ACRONYMS, AND DEFINITIONS

Terms/ Acronyms	Definition
Abantu	Abantu Environmental Services (Pty) Ltd
ACOHs	Advisory Council for Occupational Health and Safety
AEL	Atmospheric Emission Licence
Audit	Regular inspection and verification of operational activities for implementation of the approved EMPr and issued EA and AEL.
Auto-X	Auto X – P.E. Manufacturing Facility, trading as Auto-X (Pty) Ltd
BAR	Basic Assessment Report
BM Line 3	Third battery machine assembly line
BTech	Baccalaureus Technologiae
CA	Competent authority
CBA	Critical biodiversity area
CESA	Critical ecological support area
Contractor/ sub-contractor	Entity appointed on behalf of the client to install and setup/ service BM Line 3, this includes their sub-contractors and suppliers.
DEDEAT	Eastern Cape Department of Economic Development, Environmental Affairs, and Tourism
EAP	Environmental Assessment Practitioner
EAPASA	Environmental Assessment Practitioners Association of South Africa
ECA	Environmental Conservation Act 73 of 1989, as amended
ECBCP	Eastern Cape Biodiversity Conservation Plan
ECO	Environmental Control/ Compliance Officer
	A qualified person appointed by the operator or project/ site manager who will review and report on the implementation of the EMPr during the operation phase.
EMPr	Environmental Management Programme Report
	A detailed plan of action prepared to ensure that recommendations for enhancing or ensuring positive environmental impacts and limiting or preventing negative environmental impacts are implemented during the life cycle of the project. This document sets out general instructions that will be included in a contract document for the installation and setup/ service of BM Line 3. This is to ensure the operational activities are conducted and managed in an environmentally sound and responsible manner.
Environment	The surroundings within which humans exist and which are made up of the land, water, and atmosphere of the earth; micro-organisms, plant, and animal life; any part or combination of the above, and the interrelationships among and between them; and the physical, chemical, aesthetic, and cultural properties and conditions of the foregoing that influence human health and well-being.
Environmental aspect	An environmental aspect is any component of a contractor/ sub-contractor's activity that is likely to interact with and on the environment.
Environmental impact	A change in the environment, whether desirable or undesirable, that will result from the effect of an operational activity. An impact may be the direct or indirect consequence of an operational activity.
ESA	Ecological support area
General waste	Domestic waste, commercial waste, non-hazardous industrial waste, and builders' rubble e.g. paper, plastics, food, tins, wood, etc.
GHG	Greenhouse gas
GN	Government Notice

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Terms/ Acronyms	Definition
Hazardous waste	Any waste that contains organic or inorganic elements or compounds that may, owing to the inherent physical, chemical or toxicological characteristics of that waste, have a detrimental impact on health and the environment.
HEPA	High-Efficiency Particulate Air (filter)
HSA	Hazardous Substances Act 15 of 1973, as amended
Impact	A description of the potential effect or consequence of an aspect of the development on a specified component of the biophysical, social or economic environment within a defined time and space.
kg/a	Kilogram per annum
NDip	National Diploma
NEMA	National Environmental Management Act 107 of 1998, as amended
NEMAQA	National Environmental Management Air Quality Act 39 of 2004, as amended
NEMWA	National Environmental Management Waste Act 59 of 2008, as amended
NHA	National Health Act 61 of 2003, as amended
NMBM	Nelson Mandela Bay Municipality
NWA	National Water Act 36 of 1998, as amended
m	Metre
m ²	Square metre
mm	Millimetre
OHS Act	Occupational Health and Safety Act 85 of 1993, as amended
Operator or project/ site manager	A person who represents the client and is responsible for enforcing the technical and contractual requirements of the project.
PA	Protected area
Pb	Lead
Pollution	Any change in the environment caused by substances whether radioactive or other waves, or noise, odours, dust or heat emitted from any activity, including the storage or treatment of waste or substances, construction and the provision of services, whether engaged in by any person or an organ of state, where that change has an adverse effect on human health or well-being or on the composition, resilience and productivity of natural or managed ecosystems, or on materials useful to people, or will have such an effect in the future.
PPP	Personal protective equipment
<i>Pr.Sci.Nat</i>	Professional Natural Scientist
<i>Reg. EAP</i>	Registered Environmental Assessment Practitioner
Rehabilitation	The return of a disturbed area, feature or structure to a state that approximates to the state (where possible) that it was prior disruption, or to an improved state.
SACNASP	South African Council for Natural Scientific Professions
SANBI	South African National Biodiversity Institute
SANS	South African National Standards
Solid waste	Includes operational debris, chemical waste, wrapping material, timber, tins and cans, drums, wire, nails, food and domestic waste (e.g. plastic packaging and wrappers).
Waste	Any substance, whether or not that substance can be reduced, re-used, recycled and recovered • that is surplus, unwanted, rejected, discarded, abandoned or disposed of; • which the generator has no further use of for the purposes of production; • that must be treated or disposed of; or

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Terms/ Acronyms	Definition
	• that is identified as a waste by the relevant Minister by notice in the Gazette, and includes waste generated by the mining, medical or other sector, but— • a by-product is not considered waste; and • any portion of waste, once re-used, recycled and recovered, ceases to be waste.
Water pollution	Direct or indirect alteration of the physical, chemical or biological properties of a water resource so as to make it less fit for any beneficial purpose for which it may reasonably be expected to be used for, or harmful or potentially harmful to the welfare, health or safety of human beings or any aquatic or non-aquatic organisms, or resource quality, or property.
WHO	World Health Organisation

1. INTRODUCTION

1.1 PROJECT LOCATION AND DESCRIPTION

Auto X – P.E. Manufacturing Facility, trading as Auto-X (Pty) Ltd, (hereinafter referred to as Auto-X) is a holder of an Atmospheric Emission Licence (AEL), reference number 19/2/9/2/1/2/ NMBM AEL 13/36(02), in terms of section 45(1) of the National Environmental Management Air Quality Act 39 of 2004 as amended (NEMAQA) for the production (manufacturing and assembly) of lead-acid batteries. The licence was issued for lead (Pb) smelting (sub-category 4.13) and the production of acids (sub-category 7.2) as part of battery manufacturing (creation of individual components) and assembly (putting of the components together) within the Auto-X facility. The facility is situated in Korsten, Gqeberha under the Nelson Mandela Bay Municipality (NMBM) as presented in Error! Reference source not found..

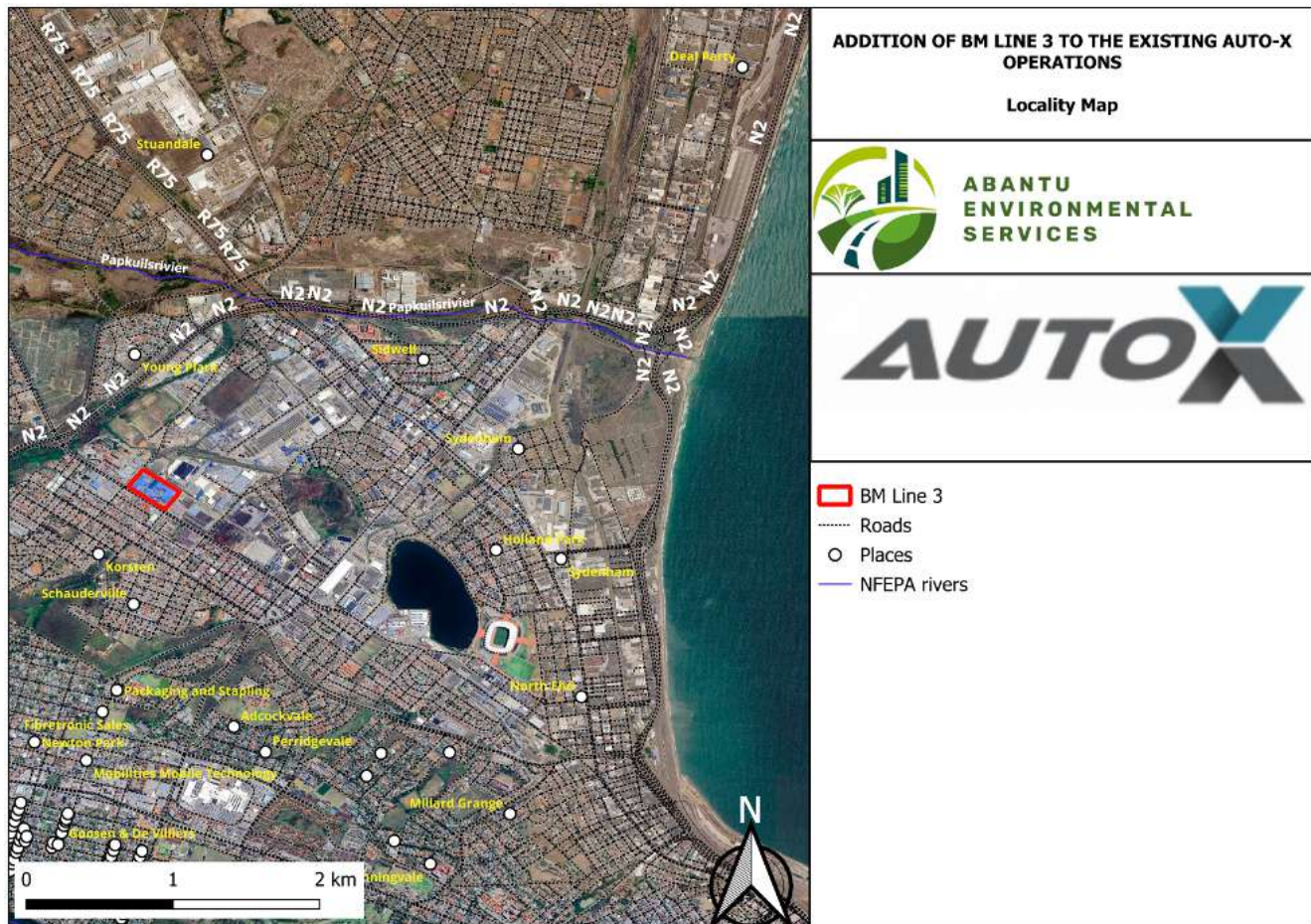


Figure 1: Locality of the existing operation and proposed addition

1.2 PROJECT DESCRIPTION

The applicant is proposing to install and operate a third battery machine line (BM Line 3) for the assembly of lead-acid batteries, along with stack AX4, at its Korsten facility. This line is an identical replica of the existing, approved BM Line 1 or 2 and will be served by a new Donaldson bag filter, identical to those already in use. This is envisaged to align with the retirement of an existing line, and planned use of latest machinery/ technology in battery assembly. Thus, alterations are to be made but installation of equipment.

Given off by the qualification that there is an existing AEL, Auto-X has an existing developed footprint, fenced off, and accessed through Lindsay and Stanford Roads. Thus, there is no construction activity associated within line addition. No other site access is necessary, as the production increase will utilise existing facilities. Existing internal roads will also continue accommodate

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traffic.

The current site measures about 55 000 square metres (m²), of which only approximately 350 m² will be utilised for the proposed installation and operation of BM Line 3.

The existing operations include two BM Lines (1 and 2) and associated infrastructure. The existing infrastructure, i.e. material storage, battery case manufacturing, acid production, Pb smelting, plate manufacturing, battery charge room, product labelling and storage, will support the proposed BM Line 3. The battery assembly comprises enveloping of positive plates and interleaving with negative plates to form elements. Elements are loaded into moulds and welded together to form cells. Cells are deposited into cases and welded together, with covers or lids heat-sealed onto the cases and end terminals burnt. The batteries are then subjected to a leak test.

The BM Line 3 will increase annual production by 220 000 automotive batteries to total 1 599 647 production capacity per annum, which remains within the maximum permitted consumption rates (2 533 450) specified in the AEL. The production increase is within limits provided for in the AEL in that the proposed increase in Pb alloy consumption (2 432 393.3 kilograms per annum [kg/a]) remains within the facility's maximum permitted consumption rate (25 998 910 kg/a).

Auto-X has lodged an AEL variation application with the NMBM and Environmental Authorisation (EA) application with the Eastern Cape Department of Economic Development, Environmental Affairs and Tourism (DEDEAT) in line with the requirements of the NEMAQA, the Environmental Impact Assessment Regulations 2014 as amended (EIA Regulations), and the National Environmental Management Act 107 of 1998 as amended (NEMA). As such, Abantu Environmental Services (Pty) Ltd (Abantu) was appointed as a Registered Environmental Assessment Practitioner (*Reg. EAP*) to lodge the EA application and conduct a Basic Assessment (BA) process for the addition of BM Line 3.

General works only include the installation, setting up, and operation of BM line 3 machinery.

1.3 APPLICABLE DOCUMENTATION

As indicated above, an AEL is in place and is subject of a variation application to align it with the proposed BM Line 3 addition. The proposed project requires EA by way of BA prior to operation of BM Line 3. This Environmental Management Programme (EMPr) must be read with the final Basic Assessment Report and applicable legislation.

The following is a list of documentation, amongst others, must be held on site and must be made available to the Environmental Control Officer (ECO), DEDEAT, and/ NMBM on request:

- Site daily diary/ instruction book/ incident reports;
- Copies of ECO reports (management and monitoring);
- EMPr;
- Complaints register;
- Method Statements;
- EA; and
- AEL.

1.4 STRUCTURE OF THE EMPr

For the holistic approach to the management of potential environmental impacts during the operation of BM Line 3, this EMPr sets out the methods by which proper environmental controls are to be implemented by the operator and other parties involved.

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This EMPr should be updated as and when required, i.e. the granting of the EA, varied AEL, etc. Any amendments made must be submitted to the DEDEAT, NMBM, ECO and project proponent/ site manager for approval prior to implementation.

1.4.1 Layout of the EMPr

The EMPr is divided into two phases, namely planning and design, and operation. Each phase has specific issues unique to that period. The potential impacts are identified and given a brief description. The phases are elaborated on as identified in the table below.

Table 1: Development phases and their descriptions

Phases	Description
Planning and design	This section will provide guidelines on planning and design activities such as complying to relevant legal, environmental, and social authorisations; continued environmental induction, training, and awareness; including health and safety.
Operation	This section will provide guideline on the practice and responsibility as required for various activities during operation.
Decommissioning and closure	This section will provide guideline on the decommissioning and closure recommendations or mitigation measures for the activities related thereto.

1.5 RECEIVING ENVIRONMENT

1.5.1 Climate

The weather pattern of the area changes throughout the day as it lies at the confluence of several climatic regimes, the most important of which are the temperate and subtropical regimes (Stone 1988). Maximum and minimum mean temperatures are experienced in February and July, respectively (McCallum 1981). Exceptionally elevated temperatures, approximately 30 degrees Celsius, can occur during berg wind conditions that develop frequently in autumn and winter. The mean annual rainfall for the Gqeberha area is approximately 600 millimetres (mm) as per Stone (1988).

Gqeberha is dominated by topographical or gradient winds for most of the year (Grobler, 2012). The area experiences westerly winds throughout the year, although in summer the percentage of easterly winds reaches more than 40% (Schumann et al, 1999). The strongest winds occur during October and November, with weakest winds during May and June (Schumann et al, 1991).

1.5.2 Topography and Hydrology

The land is relatively flat on site, occurring at an elevation range of 40 to 47 metres (m), and slopes towards the east. The proposed site drains by means of surface flow, primarily in an eastern direction towards the existing NMBM stormwater infrastructure. It is expected that groundwater flow will mimic the surface water flow.

1.5.3 Geology and Soils

In geological terms, the site is predominantly underlain by Kirkwood Formation with a little patch of Peninsula, Pakhuis, Cederberg Formations.

The Kirkwood Formation is part of the Uitenhage Group in the Eastern and Western Cape provinces, primarily within the Algoa and Gamtoos Basins. It formed during the Late Jurassic to Early Cretaceous (approximately 145 to 135 million years), along the southern margin of Gondwana (Muir et al, 2017). The formation is composed of fluvial sedimentary rocks, including variegated mudstones, medium-grained lithic sandstones (often charcoal-rich), and sporadic conglomerates (Muir et al, 2017). It is subdivided into three members, the Swartkops (estuarine sandstones), the Colchester (fossil-rich shales, siltstones, and sandstones), and the Bethelsdorp (marine microfossils in shales and sandstones) as reported by Muir et al (2017).

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The Peninsula, Pakhuis, and Cederberg Formations are part of the Cape Supergroup and are significant geological features in the Cape Peninsula (<https://vogelgat.co.za/geology/>). The Peninsula Formation Sandstone, also known as the Table Mountain Sandstone, is a key formation that has given its name to the Table Mountain Group. It consists of hard, light grey quartz arenite sandstone that is resistant to erosion (<https://vogelgat.co.za/geology/>). The Pakhuis Formation is identifiable by glacially deposited pebbles of sandstone (<https://vogelgat.co.za/geology/>). The Cederberg Formation includes dark grey siltstone and silty sandstones, which have been eroded into various structures by the wind (<https://vogelgat.co.za/geology/>).

The soil class comprise imperfectly drained soils, often shallow and often with a plinthic horizon. The soils that prevail on site have a marked clay accumulation, are strongly structured, and have a non-reddish colour. They may occur associated with one or more of vertic, melanic, and plinthic soils. Their relative wetness is favourable in dry areas, but may be seasonally wet in opposite areas.

There are no agricultural activities taking place on site, this has not been the case in more than 30 years as per available historical imagery (Google Earth, 2026). Additionally, the site has had an existing industrial facility for several years. As such, the land property for the addition of BM Line 3 is already utilised for industrial purposes.

1.5.4 Archaeology, Cultural Heritage, and Palaeontology

As indicated in subsection 1.5.3, the site can be summarised as Late Jurassic–Early Cretaceous, fluvial sediments, extremely fossil-rich, part of the Uitenhage Group. The Kirkwood Formation is the most fossil-rich unit of its age, yielding remains of theropods, sauropods, ornithopods, stegosaurs, ankylosaurs, and basal ornithomimosaurs such as *Nqwebasaurus* (Muir et al, 2017). Notable discoveries include the juvenile dryosaurid *Iyuku raathi*, *Paranthodon*, and various reptiles, amphibians, insects, mammals, and plant fossils such as ferns, conifers, cycads, and bennettitaleans (Muir et al, 2017). Mesozoic amber has also been recovered, providing insights into ancient ecosystems (Muir et al, 2017). Although the fossil-rich fossil heritage is acknowledged, the fact is that construction will not take place on site.

1.5.5 Terrestrial Biodiversity, including Animal and Plant Species

The study area is located within the Albany Thicket Biome. The site falls within the Motherwell Karroid Thicket vegetation/ ecosystem type. The vegetation/ ecosystem type is classified as Critically Endangered (GN No. 2747, 2022). Approximately 9(+3)% of the vegetation type/ ecosystem is protected, with the conservation target at 19% (South African National Biodiversity Institute [SANBI], 2018).

It should be noted that no vegetation, including animal species, occur on site and, even if they did, the proposed BM Line 3 addition does not include any construction but installation and operation of machinery on the existing warehouse currently utilised for similar activities.

The site does not occur within a terrestrial Protected Area (PA), Critical Biodiversity Area (CBA), Ecological Support Area (ESA)/ Critical Ecological Support Area (CESA), and other any other sensitive area (Eastern Cape Biodiversity Conservation Plan [ECBCP], 2019; NMBM, 2009).

1.5.6 Aquatic Biodiversity

The proposed development falls within the Mzimvubu to Tsitsikamma Water Management Area. At a desktop level, the site constitutes aquatic ESA 1 (SANBI, n.d; ECBCP, 2019). The ESAs are vital regions that support the functioning of PAs and CBAs, playing a crucial role in delivering ecosystem services and enhancing landscape connectivity (SANBI, 2018). The ESAs are defined as areas in a natural, near natural, or moderately degraded condition that are not essential for meeting biodiversity targets but are important for supporting the functioning of the areas referred to above. They are often vital for delivering ecosystem services, which include benefits such as clean water, climate regulation, and habitat for wildlife (SANBI, 2018). As per the site investigation conducted in June 2026, the site is in a developed state and no longer displays the sensitivity referred

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to.

Rivers, Papkuils and Papenkuils, around the general area occur more than 175 m from the site. The National Freshwater Ecosystem Priority Area wetlands are located more than 750 m from the site.

The majority of the site extent has hardened surfaces which drain to the existing stormwater infrastructure during rainfall events.

1.5.7 Site Sensitivity

The proposed BM Line 3 addition is to be installed and operated in an existing industrial facility already set up for similar activities. Thus, there will be no construction associated with the installation and operation. The site does not occur within a terrestrial PA, CBA, ESA/ CESA, and other any other sensitive area. As such, any sensitive environments occurring around the site (**Figure 2**) are not anticipated to be impacted on by the proposed BM Line 3 addition.



Figure 2: Biodiversity sensitivity around the site (none on site)

2. PURPOSE OF THE ENVIRONMENTAL MANAGEMENT PROGRAMME

The EMPr forms part of the contractual obligations to which all operators/ employees involved in planning and design, operation and maintenance, including decommissioning and closure work must be committed to. It serves as a guideline and baseline information document for the construction and operations of the proposed project and aims to comply with section 24N of the NEMA as well as the EIA Regulations.

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This section outlines the environmental management guideline measures as encompassed in **Appendix 4** of the EIA Regulations that will be undertaken to avoid or minimize potential impacts on the environment throughout all phases of the proposed project.

The content of the EMPr, prepared in terms of the EIA Regulations, is outlined in **Table 2** below. The relevant sections of the EMPr, in which the requirements are addressed, is also outlined.

Table 2: Alignment to Appendix 4 of the EIA Regulations

Prescribed content of an EMPr	Reference in the EMPr
1. (1) An EMPr must comply with section 24N of the Act and include- (a) details of – (i) the Environmental Assessment Practitioner (EAP) who prepared the EMPr. (ii) the expertise of that EAP to prepare an EMPr, including curriculum vitae.	Page ii and iv, CV attached to the BAR
(b) a detailed description of the aspects of the activity that are covered by the EMPr as identified by the project description.	Section 1.1, 1.2, and Error! Reference source not found.
(c) a map at an appropriate scale which superimposes the proposed activity, its associated structures, and infrastructure on the environmental sensitivities of the preferred site, indicating any areas that should be avoided, including buffers.	Section 1.5.7 and Figure 2
(d) a description of the impact management outcomes, including management statements, identifying the impacts and risks that need to be avoided, managed, and mitigated as identified through the environmental impact assessment process for all phases of the development including- (i) planning and design. (ii) pre construction activities. (iii) construction activities. (iv) rehabilitation of the environment after construction and where applicable post closure. (v) where relevant, operation activities.	Section 5, 6, 7 & Section 8
(f) A description of proposed impact management actions, identifying the manner in which the impact management outcomes contemplated in paragraphs (d) will be achieved, and must, where applicable, include actions to – (i) Avoid, modify, remedy, control or stop any action, activity or process which causes pollution or environmental degradation. (ii) Comply with any prescribed environmental management standards and practices. (iii) Comply with any applicable provisions of the Act regarding closure, where applicable. (iv) Comply with any provisions of the Act regarding financial provisions for rehabilitation, where applicable.	Section 5 to 5.2
(g) The method of monitoring the implementation of the impact management actions contemplated in paragraph (f).	Section 5 to 5.2
(h) The frequency of monitoring the implementation of the impact management actions contemplated in paragraph (f).	Section 5 to 5.2
(i) An indication of the persons who will be responsible for the implementation of the impact management actions.	Section 6
(j) The time periods within which the impact management actions contemplated in paragraph (f) must be implemented.	Section 5 to 5.2
(k) The mechanism for monitoring compliance with the impact management actions contemplated in paragraph (f).	Section 7
(l) A program for reporting on compliance, taking into account the requirements as prescribed by the Regulations.	Section 7

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Prescribed content of an EMPr	Reference in the EMPr
(m) An environmental awareness plan describing the manner in which – (i) The applicant intends to inform his or her employees of any environmental risk which may result from their work. (ii) Risks must be dealt with in order to avoid pollution or the degradation of the environment.	Section 3.2, 6, and 7
(n) Any specific information that may be required by the competent authority.	None requested
1. (2) Where a government notice gazetted by the Minister provides for a generic EMPr, such generic EMPr as indicated in such notice will apply.	Not applicable

The main objectives of this EMPr are therefore to:

- Outline environmental management measures related to project activities and provide project operators and/ employees with guidelines for carrying out activities in such a way as to minimise environmental impacts;
- Be used as a foundation for the specific environmental management instructions contained in operation contract documents, where compliance will be a contractual obligation for operators and/ employees;
- Be used as an educational tool, for orientation and training of project personnel and contractors;
- To outline functions and responsibilities of responsible persons;
- To state standards and guidelines, which are required to be achieved in terms of environmental legislation;
- To outline mitigation measures and environmental specifications which are required to be implemented for all phases of the project in order to minimise the extent of environmental impacts, and to manage potential environmental impacts associated with the proposed project; and
- To prevent long-term or permanent environmental degradation.

2.1 ENVIRONMENTAL CODE OF CONDUCT

One of the objectives of the EMPr is to ensure that all the workforce, operator, sub-contractors, and any other parties involved understand environmental issues and potential impacts related to the site activities. This environmental code of conduct provides the basic rule that must be strictly adhered to. It is the responsibility of the operator or site environmental officer and the ECO, where appointed, to ensure that each contractor, sub-contractor, and workforce understand and adhere to this code of conduct. All persons are obliged to keep to the rules of this code of conduct. Ignorance, negligence, recklessness, or a general lack of commitment resulting in environmental degradation or pollution must not be tolerated.

Environmental rules, amongst others, include:

- Do not waste electricity, water, or consumables;
- Only use authorised accesses;
- Do not litter;
- Dispose solid waste into the correct waste containers provided;
- Prevent pollution;
- Use the toilet facilities provided;
- Do not dispose of contaminated wastewater to the stormwater channels or the environment;
- Immediately report any spillage from containers, plant, or vehicles;
- Do not burn or bury any waste on site;
- Do not trespass onto private properties;
- Know the firefighting procedure and locations of firefighting equipment; and
- Know the environmental incident procedures.

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2.2 CONTRACTUAL OBLIGATION

This EMPr must be included in all contract documentation associated with BM Line 3 installation and operation. Note that, notwithstanding the Companies Act 71 of 2008, as amended, or the Close Corporations Act 69 of 1984, as amended, the directors of a company or members of a close corporation are jointly and severally liable for any negative impact on the environment, whether advertently or inadvertently caused by the company or close corporation which they represent, including damage, degradation or pollution.

3. MANAGEMENT AND MONITORING PROCEDURE

3.1 ORGANISATIONAL STRUCTURE AND RESPONSIBILITIES

This section indicates the parties responsible for implementing the environmental measures and action plans laid out in this EMPr. Formal responsibilities are necessary to ensure that key procedures are executed. Specific responsibilities of the project proponent/ site manager (Auto-X), operator/ sub-contractor (personnel responsible for the installation and setup of BM Line 3, if not Auto-X), and ECO (where appointed) are as detailed below.

Project proponent/ site manager shall:

- Be fully acquainted with the EMPr;
- Ensure that the contractor/ sub-contractor is aware of all specifications, legal constraints, standards, and procedures pertaining to the project specifically with regard to the environment;
- Ensure that all stipulations within the EMPr are communicated and adhered to by the contractor/ sub-contractor;
- Monitor the implementation of the EMPr throughout the operations by means of regular site visits and meetings; and
- Order the removal of any person(s) and/or equipment in contravention of the specifications of the EMPr.

The contractor/ sub-contractor shall:

- Be fully conversant with the EMPr;
- Ensure compliance with the EMPr;
- Ensure that all the environmental specifications contained within this EMPr are adhered to; and
- Regularly liaise with the project manager/ site manager on matters relating to the environment.

The above responsibilities listed for the contractor/ sub-contractor will also apply to any appointed sub-consultants.

The ECO shall:

- Be fully appraised with the EMPr;
- Be fully conversant with all environmental legislation and ensure compliance;
- Ensure that all the environmental specifications contained within this EMPr are adhered to;
- Regularly liaise with the project proponent/ site manager on matters relating to the environment; and
- Compile reports as to the compliance of operations with the conditions of any applicable approvals and requirements of this EMPr, and report to all parties involved (project proponent/ site manager).

3.2 ENVIRONMENTAL AWARENESS TRAINING

Training and environmental awareness is an integral part of an EMPr. The overall aim of the training will be to ensure that all site staff is informed of their relevant requirements and obligations pertaining to the relevant authorisations, licences, permit,

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and the approved EMPr for the protection of the environment. Environmental awareness training courses should be run for all personnel on site, whether permanently or temporary visiting the site. Two types of courses should be run, one for the project proponent's/ site manager's or contractor's/ sub-contractor's management and one for all site staff and labourers. Courses shall be run in the morning during normal working hours, at a suitable venue provided by the project proponent/ site manager. All attendees shall remain for the duration of the course and sign an attendance register on completion that clearly indicates participant's names, a copy of which shall be handed to the ECO.

The size of each session shall be limited to 30 people. The project proponent/ site manager or contractor/ sub-contractor shall allow for sufficient sessions to train all personnel. Subsequent sessions shall be run for any new personnel coming onto site. A method statement with respect to the organisation of these courses shall be submitted to the ECO.

Notwithstanding the specific provisions of this clause, it is incumbent upon the project proponent/ site manager or contractor/ sub-contractor to convey the sentiments of the EMPr to all personnel involved with the operations.

3.2.1 Training Course for Management and Foremen

The environmental awareness training course for management shall include all management staff and foremen. The course, which will be presented by the site environmental officer or ECO (where appointed), will be of approximately one-hour duration.

The initial course shall be undertaken not less than seven days prior to commencement of the works. Subsequent courses shall be held as and when required.

3.2.2 Training Course for Site Staff and Labour

The environmental awareness training course for site staff and labour shall be presented by the site manager from material provided by the ECO or the site environmental officer, unless otherwise required by the project specification. The course will be approximately one-hour long.

The course shall be run not more than seven days after commencement of the works with sufficient sessions to accommodate all available personnel. Subsequent courses shall be held as and when required.

3.2.3 Information Posters

The project proponent/ site manager shall erect and maintain information posters for the information of the employees depicting actions to be taken to ensure compliance with the EMPr. Information posters shall be laminated and erected in all eating areas, workshops, and site offices. The project proponent/ site manager shall ensure that the information posters are not damaged in any way and shall replace them if any part becomes illegible.

3.2.4 Environmental Awareness Training Programme

The environmental awareness training programme should include:

- The induction of all operation staff;
- Signing by all persons of an acknowledgement of receiving and understanding the induction;
- Identification of environmental risks and job specific training on addressing these risks; and
- Training on the implementation of emergency procedures.

3.2.5 Topics Covered by the Environmental Awareness Programme

Various topics to be entailed in environmental awareness programme include, but are not limited to:

- What is meant by "Environment"?

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- Why does the environment need to be protected and conserved?
- How can operational activities impact on the environment, i.e. dumping of hazardous waste in stormwater channels?
- What can be done to mitigate against such impacts?
- Awareness of emergency and spill response provisions.

4. LEGISLATION AND GUIDELINES

All the applicable environmental standards contained within the applicable environmental legislation must be adhered to. Below are applicable legislation and guidelines pertaining to the proposed BM Line 3 addition. The process is undertaken as part of the impact assessment for the proposed BM Line 3 addition as part of the BA process due to the triggered activities in terms of Listing Notice 1 of the EIA Regulations, including Government Notice (GN) No. 893 of 2013 as amended (Listed Activities and associated Minimum Emission Standards) enacted in terms of the NEMAQA. An EA and AEL variation applications are therefore being undertaken.

4.1 NATIONAL ENVIRONMENTAL MANAGEMENT ACT 107 OF 1998, AS AMENDED

The NEMA is a 'principles-based act' and is an overarching statute regulating various aspects of natural resource use, integrated environmental management, and pollution control. The legislation gives effect to section 24 of the Constitution of the Republic of South Africa, 1996, regarding the right to an environment that is not harmful to the health and wellbeing of anyone within the borders of South Africa. The objectives of the framework environmental law include sustainable development, environmental protection, equitable distribution of natural resources, and the formulation of environmental management frameworks, amongst others.

The statute aims to provide for cooperative environmental governance by establishing principles for decision making on matters affecting the environment, institutions that will promote co-operative governance, and procedures for coordinating environmental functions exercised by organs of state. Thus, although the AEL variation application is governed by the NMBM, the DEDEAT in their administering of the EA application are also coordinating with the NMBM in this regard.

Section 24 provides for the prohibition, restriction, and control of activities which are likely to have a detrimental effect on the environment. Auto-X is committed to continue managing potential environmental impacts as they have been doing throughout the operational phase of their facility.

4.2 NATIONAL ENVIRONMENTAL MANAGEMENT AIR QUALITY ACT 39 OF 2004, AS AMENDED

The purpose of the NEMAQA is to reform South Africa's air-quality law by introducing measures to prevent pollution, protect the environment, and ensure sustainable development through national norms and standards for air-quality monitoring, management, and control.

The prescription aims to protect the environment by preventing air pollution and ecological degradation; secure ecologically sustainable development while balancing economic and social development; establish national norms and standards for air-quality monitoring, management, and control across all spheres of government; and introduce specific air-quality measures to improve ambient air quality.

The statute's preamble further emphasises the need to address poor ambient air quality in many parts of South Africa; the disproportionate health burden on poorer communities caused by polluted air; the environmental and economic costs of air pollution, often not borne by the polluter; the harmful effects of ozone-depleting substances and greenhouse gases; and the constitutional right to an environment not harmful to health or well-being.

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The act also provides for a national air-quality framework guiding all levels of government; ambient air-quality and emission standards at national, provincial, and local levels; air-quality management plans integrated into broader environmental planning; and licencing of listed activities that may impact air quality. The existing AEL and continual compliance with the requirements of the NEMAQA will be upheld by the applicant.

4.3 NATIONAL ENVIRONMENTAL MANAGEMENT WASTE ACT 59 OF 2008, AS AMENDED

The National Environmental Management Waste Act 59 of 2008, as amended (NEMWA) regulates waste management in South Africa to protect health and the environment, promote sustainable development, and establish national standards and licencing for waste activities.

The NEMWA aims to reform waste management laws to prevent pollution and ecological degradation, ensure ecologically sustainable development, and protect public health and the environment. It emphasises the minimisation, re-use, recycling, and recovery of waste, treating disposal as a last resort. The statute recognises waste as a potential resource with economic opportunities.

The act applies to all waste generators, transporters, and disposal facilities in South Africa, except for certain exclusions. It covers domestic, business, building and demolition, hazardous, and inert waste.

Key provisions:

- Integrated Waste Management Plans: Required for organs of state and industries to ensure coordinated waste management.
- Licencing and control: Waste management activities, including generation, storage, transport, treatment, and disposal, require licences issued by designated authorities.
- Extended Producer Responsibility: Producers may be required to manage the post-consumer stage of their products, including recycling and awareness programs.
- Priority Wastes: Certain hazardous or high-risk wastes can be declared as priority, triggering specific management and remediation obligations.
- Compliance and enforcement: Non-compliance is a criminal offense, with penalties ranging from fines to imprisonment, and may include remedial orders to repair environmental damage.
- National Waste Information System: Establishes a system for monitoring and reporting on waste management activities across all spheres of government.
- Remediation of contaminated land: Provides for investigation, assessment, and remediation of contaminated sites to protect health and the environment.

As is the case with other legislation, the act promotes cooperative governance, requiring national norms and standards to be applied uniformly across provinces. It encourages clean production, waste minimisation, and environmentally sound management practices. Municipalities retain certain powers to regulate local waste disposal, but must align with national standards.

In essence, the NEMWA establishes a comprehensive legal framework for sustainable waste management, balancing environmental protection, public health, and economic considerations. It sets out responsibilities for all waste stakeholders, enforces compliance, and promotes the circular economy through recycling, recovery, and producer responsibility measures.

Auto-X has an established system for the handling and management of all waste produced with the facility, this practice will continue throughout the operations.

4.4 NATIONAL WATER ACT 36 OF 1998, AS AMENDED

The National Water Act 36 of 1998, as amended (NWA), aims to provide for fundamental reform of the law relating to water resources, to repeal certain laws, and to provide for matters connected therewith. It establishes suitable institutions and ensures appropriate community, racial, and gender representation. The NWA is designed to protect, conserve, and manage water resources in a sustainable and equitable way, promoting social and economic development through the use of water. It also mandates equitable access to water, particularly for disadvantaged communities, and acknowledges the economic value of water. The NWA is committed to the protection of water resources from pollution and degradation, maintaining ecological integrity, and fostering public participation in water resource management.

4.5 HAZARDOUS SUBSTANCE ACT 15 OF 1973, AS AMENDED

The Hazardous Substances Act 15 of 1973, as amended (HSA), regulates the control, import, manufacture, sale, and use of substances that may cause injury, ill-health, or death due to their toxic, corrosive, irritant, sensitising, flammable, or pressure-generating properties. The act was enacted to protect public health and safety by controlling substances that pose significant risks to humans. It also extends to certain electronic products that may present hazards. The statute provides a legal framework for classification, prohibition, and regulation of hazardous substances.

Substances, within the act, are divided into groups based on the degree of danger they present. These groups determine the level of control, licencing requirements, and restrictions on sale, use, or disposal. Group IV substances, for example, are subject to strict production, acquisition, disposal, import, and export controls. The HSA requires licencing for the manufacture, import, export, and use of certain hazardous substances.

Authorised inspectors are empowered to enforce the statute, including powers of embargo, seizure, and examination of substances. Analysts may conduct further analysis of samples to ensure compliance. The legislation also outlines liabilities, special defences, warranties, and employer responsibilities in relation to hazardous substances.

The act specifies offences and penalties for non-compliance, including illegal importation, manufacture, sale, or use of hazardous substances. It also provides for forfeiture and disposal of goods and sets out time limits and procedural requirements for prosecution.

The HSA addresses its application to substances in transit and its operation in relation to other laws. Regulations under the act provide detailed guidance on compliance and safety measures.

In summary, the HSA establishes a comprehensive legal framework to protect public health and safety by regulating hazardous substances through classification, licencing, inspection, and enforcement mechanisms.

4.6 OCCUPATIONAL HEALTH AND SAFETY ACT 85 OF 1993, AS AMENDED

The Occupational Health and Safety Act 85 of 1993, as amended (OHSA) regulates workplace health and safety, protects employees, and others from hazards, and establishes structures such as the Advisory Council for Occupational Health and Safety (ACOHS).

The OHSA aims to ensure safe and healthy working environments across South Africa. It applies to nearly all workplaces and places duties on employers, employees, manufacturers, and others involved in workplace activities.

Some of the key objectives of the statute include to protect the health and safety of persons at work, ensure safety in connection with the use of plant and machinery, safeguard non-employees from hazards arising from workplace activities, and to establish the ACOHS.

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The employers are required to provide and maintain a safe working environment, identify hazards and assess risks, inform employees of workplace dangers, and ensure proper training and supervision, among other things. At the same time, employees must take reasonable care for their own safety, comply with safety instructions, report unsafe conditions, and avoid interfering with safety equipment, inter alia.

Manufacturers and suppliers must ensure that articles and substances used at work are safe and without risk when properly used.

It is a requirement under the act that health and safety representatives be appointed, health and safety committees be established, and functions be defined for both representatives and committees.

Inspectors, provided for under the legislation, have powers to enter workplaces, conduct investigations, and issue compliance notices. The OHSA outlines penalties for non-compliance, including fines and imprisonment for serious violations. The statute remains the foundational legislation governing workplace health and safety.

General Administrative Regulations, 2003 as amended, enacted under the OHSA describe the administration of the various OHS Regulations, including the reporting and recording of incidents and occupational diseases. This is applicable to operators, contractors/ sub-contractors during the planning, design, operational, and decommissioning phases of the project.

Auto-X has and will continue to comply with the provisions of this act and implement and communicate all the necessary measures to keep everyone within the site safe.

4.7 NATIONAL HEALTH ACT 61 OF 2003, AS AMENDED

The National Health Act 61 of 2003, as amended (NHA), establishes a framework for a uniform, equitable health system, defining the rights and responsibilities of users, providers, and government bodies. The Act aims to create a structured national health system that integrates public and private health services, ensuring equitable access to healthcare across the country while addressing historical disparities. Its objectives include:

- Establishing a national health system encompassing all health service providers.
- Defining the rights and duties of healthcare providers, health workers, health establishments, and users.
- Protecting and promoting the constitutional right to access healthcare, including reproductive health services, and safeguarding vulnerable groups such as women, children, older persons, and persons with disabilities.
- Ensuring the provision of essential health services, particularly primary healthcare, within available resources.
- Promoting equity in healthcare delivery and prioritising services based on population needs and resource availability.

The NHA guarantees that health service users have informed consent for medical procedures, access to emergency treatment without delay, the right to participate in decisions regarding their healthcare, and protection against treatment without consent, except in emergencies or as authorised by law.

The act allows the Minister of Health, in consultation with the Minister of Finance, to prescribe conditions for free health services. Key provisions include:

- Free services for pregnant and lactating women and children under six not covered by medical aid.
- Free primary healthcare services for all persons not covered by medical aid or compensation schemes.
- Free termination of pregnancy services in accordance with the Choice on Termination of Pregnancy Act 92 of 1996.

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The NHA thus provides a comprehensive legal framework to ensure equitable, ethical, and efficient healthcare delivery, balancing the rights of users with the responsibilities of providers and government authorities.

4.8 NATIONAL NOISE CONTROL REGULATIONS 1992, AS AMENDED

The National Noise Control Regulations 1992, as amended (GN R154 of 1992), sets out the legal standards for controlling disturbing noise, define local authority powers, and establish prohibitions and exemptions under section 25 of the Environment Conservation Act 73 of 1989, as amended (ECA). The GN R154 of 1992 remain in effect despite the ECA being largely replaced by the NEMA. These regulations provide a framework for controlling noise pollution, defining what constitutes a disturbing noise and outlining the responsibilities of individuals, businesses, and local authorities.

Local authorities are empowered to conduct inspections or tests on vehicles, machinery, or premises to assess noise levels; enforce compliance with noise limits and issue directives to reduce or eliminate disturbing noise; and require Noise Impact Assessments for activities likely to generate significant noise.

No person may make, produce, or cause a disturbing noise, nor allow it to be caused by any machine, device, or combination thereof. The regulations apply broadly to industrial, commercial, and public activities, but certain exemptions exist. Noise regulations do not apply in cases where sound is emitted to warn people of a dangerous situation and when sound is produced during an emergency.

Noise levels are measured using approved instruments and compared against zone sound levels or ambient levels. Although the sound level emitted within the facility do not within the ambits of these regulations, care is still taken within the site to protect hearing of employees and operators by wearing ear blockers in areas of elevated sound levels.

The regulations establish a legal framework to prevent and manage noise pollution, empower local authorities to enforce limits, and define clear standards for what constitutes disturbing noise. Compliance involves monitoring, assessment, and, where necessary, mitigation measures, with exemptions for safety and emergency situations. These regulations are complemented by provincial rules and local enforcement practices, ensuring a comprehensive approach to noise management.

4.9 OTHER LEGISLATION OR GUIDELINES CONSIDERED

- South African National Standard (SANS) 10103:2008 (The Measurement and Rating of Environmental Noise with Respect to Annoyance and Speech Communication)
- SANS 1929:2005 (Ambient Air Quality Management)
- NMBM Air Pollution Control By-Law
- NMBM Integrated Development Plan
- NMBM By-Laws (Waste Management, Solid Waste, Water and Sanitation)

5. ENVIRONMENTAL MANAGEMENT REQUIREMENTS

5.1 PLANNING AND DESIGN

This section of the EMPr provides management principles and mitigation measures for the pre-operational phase of the BM Line 3. The environmental actions, procedures, and responsibilities as required during this phase are specified below.

Table 3: Impacts and Mitigation Measures in Planning and Design Phase

	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1.	Permitting requirements	Make all necessary arrangements with applicable stakeholders to ascertain that all new line requirements are met.	Operator/ site manager	Once-off
2.	Energy efficient machinery	- Ensure the battery assembly machinery technology is energy efficient than the existing machinery. - By prioritising energy efficiency in battery assembly, manufacturers can achieve both economic and environmental benefits, contributing to a more sustainable energy future. - Energy-efficient processes minimise the amount of energy used during the production of batteries, leading to lower operational costs.	Operator/ site manager	Once-off
3.	Job opportunities	Ensure any new job opportunities accrue to the local community.	Operator	Once-off/ as and when required
4.	Environmental awareness, roles and responsibilities and implementation	- The EMPr should be included as part of the contract or documentation thereby making it part of the enquiry document to make the recommendations and constraints, as set out in this document, enforceable under the general conditions of contract. - A copy of this EMPr should be available on site. The operator must ensure that all the personnel on site, contractors/ sub-contractors and their team, suppliers, etc. are familiar with and understand the specifications contained in the EMPr. - Before construction activities commence, role players must have a clear indication of their role in the implementation of this EMPr.	Operator/ site manager	Once-off
5.	Pro-forma documentation	- The contact details of the ECO, contractor, and SHE (Safety, Health, and Environmental) officer should be completed as part of the pro-forma documents and a copy must be kept on site. This document should be made available to the approving authority on request. - Subcontractor(s) contracts with the principle contractor must contain a clause to the effect that the disposal of all packing refuse/ waste to an officially approved dumping site is the responsibility of the subcontractor in question and that the subcontractors are bound to the management activities stipulated in this EMPr.	Operator/ site manager	Once-off
6.	Method statements	Certain method statements should be provided by the contractor. All activities which require method statements may only commence once the method	Operator/ site manager and or ECO	As and when required

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ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
	statements have been approved by the operator/ site manager and or ECO, as applicable. Where applicable, the contractor will provide job-specific training on an ad hoc basis when workers are engaged in activities, which require method statements.		

5.2 OPERATIONAL PHASE

This section of the EMPr provides management principles for the operational phase of BM Line 3. Environmental actions, procedures, and responsibilities as required during the operational phase are specified.

Table 4: Impact and Mitigation Measures in Operational Phase

ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1. - Release of Pb and H₂SO₄ into the environment - Release of fumes/ air emissions due to the welding/ burning associated with the battery assembly - Environmental pollution, including air pollution, and health risks associated with Pb, H₂SO₄, and fumes	- Minimise the emissions of Pb, H₂SO₄, and any other materials utilised within the facility. - Regular testing/ sampling of Pb levels to ascertain emissions are minimised. Periodic ambient Pb monitoring must be considered in the easterly corner of the facility or neighbouring property to verify Pb emissions against the National Ambient Air Quality Standards published under Government Notice 1210 of 2009. - Duct audit should be undertaken frequently. - Immediate action following any accidental release of H₂SO₄ involves containment to prevent further spread and neutralisation to mitigate corrosive effects, often employing alkaline agents such as lime or soda ash. - Long-term action necessitates the implementation of improved safety protocols, including enhanced storage facility design, employee training, and emergency response drills. - Thorough analysis of H₂SO₄ release incidents is essential for identifying root causes, evaluating the effectiveness of response measures, and informing preventative strategies. - Continue with the monitoring of air quality, correct personal protective equipment (PPE) use, and implementing air pollution abatement measures. - Continuously engage with employees on the safety and health issues, i.e. the current use imagery around the facility and the toolbox talks. - Strengthen policies and plans to prevent Pb exposure, including the adoption of standards and strong implementation of these measures. - Continue to educate and provide refresher talks on safely handling of the materials used on the facility such that that do not become a danger to the workers. - Follow the World Health Organisation (WHO) developments on any initiatives and further measures for safely working with Pb.	Operator/ site manager/ environmental officer/ appointed ECO	Ongoing

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	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		Conduct regular inspections using emissions leak detection technology.		
2.	Energy consumption	- Continue to apply the energy saving measures discussed above, i.e. insulated chilled water pipelines to keep the water cool such that less energy is required for cooling; use of energy efficient generators; light-emitting diodes/ LED; supplementing the NMBM provided electricity with Liquified Petroleum Gas, etc. - Optimise industrial processes, where feasible, and implementing smart technologies for demand management. - Conduct comprehensive energy audits to identify high-consumption areas and implement targeted efficiency upgrades. - Encourage management and workers behaviour changes and appliance upgrades to minimise demand. - Ensure regular servicing, inspection, and upgrading of machinery for efficiency. - Continue implementing energy saving measurements.	Operator/ site manager/ environmental officer/ appointed ECO	Ongoing
3.	GHG emissions	- Continue with the monitoring of greenhouse gas (GHG) emissions. - Effective mitigation results in verifiable reductions in operational costs and associated GHGs, improving corporate sustainability metrics. - Continue to conduct comprehensive lifecycle assessments to quantify and mitigate GHG footprints associated with battery production, including material sourcing and manufacturing. - Continue with extended producer responsibility practices implemented. - Establish clear supply chain transparency and ensuring adherence to environmental and labour standards throughout the entire value chain.	Operator/ site manager/ environmental officer/ appointed ECO	Ongoing
4.	- Job retention - Skills development and transfer - Employment creation and skills development associated with battery recycling and other second-life applications	- Foster skilled jobs in battery recycling and second-life applications, which can positively impact local economies. - Continue to enhance local manufacturing capacity, job creation and retention, while also supporting the energy value chain.	Operator/ site manager/ environmental officer	Ongoing
5.	Localised, around some machinery, elevated noise levels	- Continue with the use of correct PPE. - Establish a long-term maintenance and monitoring plan, including budget allocation. - Continue with the training of operators on rapid incident response and reporting protocols.	Operator/ site manager/ environmental officer	Ongoing
6.	Production of H₂SO₄ can cause acid rain which has a negative effect to	- Continue with the developed implementation of scheduled preventative maintenance programme. - Consider closed-loop recycling, where feasible. This involves processing used lead-acid batteries to extract Pb and H₂SO₄, using these materials to manufacture	Operator/ site manager/ environmental officer	Ongoing

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	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
	soil, plants, and aquatic life - The mining of Pb can result in environmental degradation - Depletion of finite resources, i.e. Pb alloy	new batteries. It lessens the impact on the environment and the need for additional raw resources. - Ongoing research focuses on developing advanced recycling technologies that are more energy-efficient and environmentally friendly. Innovations in hydrometallurgical and pyrometallurgical processes aim to enhance recycling efficiency and minimise emissions. - Promote responsible disposal practices to minimise environmental harm. - Continue to adhere to policies and regulations governing the proper disposal and recycling of lead-acid batteries. - Implement risk mitigation strategies, such as proper storage, transportation, and treatment of spent batteries, along with public education and awareness campaigns. This helps minimise environmental risks and ensure responsible battery management. - Adhere to the EMPr and ensure that contractors are trained in environmentally responsible practices. - Develop robust recycling infrastructure and promote the circular economy to manage battery waste effectively. - Continue harvesting rainwater and implementing water saving measures.		

5.3 DECOMMISSIONING AND CLOSURE PHASE

This section of the EMPr provides management principles for the decommissioning and closure phase of BM Line 3. Environmental actions, procedures, and responsibilities as required during this phase are specified.

Table 5: Impact and Mitigation Measures in Closure and Decommissioning Phase

	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
1.	- Job losses - Skills and skills transfer loss - Negative effect in the lead-acid battery market	- Develop a formal workforce-transition plan well in advance of closure. Provide affected employees with written notice, information on severance benefits, pension/provident-fund arrangements and their legal rights. - Provide affected workers with financial counselling, budgeting assistance, and debt-management support. Where feasible, phase workforce reductions to give employees time to seek alternative employment. - Implement retraining and skills-development programmes targeted at affected workers, including training in electrical work, mechanical maintenance, waste management, environmental monitoring, logistics, construction and other locally marketable skills. - Prioritise local recruitment for decommissioning, demolition, transportation, rehabilitation, security, landscaping and environmental-monitoring activities,	Operator	Ongoing, post closure

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	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		provided workers meet competency requirements. • Where possible, use local businesses for decommissioning-related procurement. • Develop a local economic-transition strategy, potentially including support for small businesses, enterprise development, skills programmes and linkages with other employers in the area. • Engage the relevant municipality regarding anticipated changes in the site's economic contribution and investigate opportunities for alternative productive reuse of the site. • Establish a structured stakeholder engagement programme. Clearly communicate the closure schedule, reasons for closure, employment implications, environmental risks, rehabilitation activities and future land-use plans. • Establish a transparent grievance mechanism through which employees, contractors, and community members can submit complaints and receive documented responses. • Identify social investments previously supported by the facility and develop a transition plan with beneficiaries, rather than abruptly terminating critical programmes. • Provide specialised training, PPE, medical surveillance, safe-work procedures and competent supervision for workers potentially exposed to Pb, H₂SO₄, contaminated dust, residues, and other hazardous materials. • Identify informal traders, service providers, and other vulnerable groups that may have benefited from the facility and consider appropriate transition measures. • Communicate closure decisions honestly and consistently, including the expected duration, employment measures, environmental safeguards, and proposed future use of the site. • Conduct a vulnerability assessment and give particular attention to low-income households, single-income households, and workers with limited transferable skills. • Identify local suppliers and contractors dependent on the facility and provide advance information about closure.		
2.	Lessened environmental pollution, including air pollution Free up of energy for other uses	• Ensure that adequate financial provision is available for decommissioning, rehabilitation, and post-closure obligations. South African environmental legislation provides for financial provision for specified rehabilitation, decommissioning, closure and post-closure activities. • Isolate all sources of electrical and mechanical energy to ensure safety. • Physically disconnecting battery modules and other components. • Package batteries in United Nations-approved, non-conductive containers with adequate padding and following hazardous material documentation. • Direct recycling to recover valuable materials or repurposing for less demanding applications.	• Operator	• Ongoing, post closure

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	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		• Adhering to local, regional, and national regulations for waste handling and disposal. • A formal decommissioning process is crucial for ensuring safety, regulatory compliance, and environmental responsibility. It standardises procedures across multiple sites, minimises liability risks, maximises resource recovery through recycling, and provides clear accountability at each stage from disconnection to final disposition or second-life use. • Conduct a detailed site inspection and environmental due-diligence assessment before dismantling. Identify areas where Pb compounds, PbO, Pb paste, H₂SO₄, electrolyte, contaminated sludge, batteries, chemicals and wastes were handled. Review historical spills, waste-disposal areas, drainage systems and underground infrastructure. Establish baseline soil, groundwater, surface-water and, where relevant, indoor/ outdoor dust conditions. • Survey floors, walls, roofs, machinery, ventilation systems, drains and process equipment for Pb contamination. Establish controlled work zones. Use methods that minimise dust generation during dismantling. Avoid dry sweeping and uncontrolled compressed-air cleaning; use wet cleaning or suitable High-Efficiency Particulate Air (HEPA) filtered industrial vacuum systems. • Inventory all finished, defective, partially assembled and waste batteries. Segregate batteries from other wastes and store them in a secure, covered and appropriately contained area. Prevent physical damage, short-circuiting and leakage. Send batteries to appropriately authorised recycling/ recovery facilities rather than ordinary landfill disposal. Lead-acid batteries should generally be recycled because improper disposal can release both Pb and H₂SO₄. • Drain, collect and manage residual electrolyte only using controlled procedures and suitable acid-resistant equipment. Do not discharge electrolyte to soil, stormwater drains or ordinary wastewater systems. Store recovered acid in compatible, labelled and banded containers. Neutralisation or recovery should be undertaken by competent personnel using an approved procedure. • Implement dust suppression during dismantling, demolition and material handling. Use wet methods where appropriate, local exhaust ventilation and HEPA filtration. Enclose particularly contaminated work areas. Keep potentially contaminated surfaces damp during disturbance and prevent windblown dust. WHO identifies Pb dust and particles as important exposure pathways during battery handling and recycling. • Decontaminate machinery, conveyors, mixers, casting equipment, charging equipment, ventilation ducts, floors and other process equipment before removal from controlled areas. Characterise residues generated during cleaning and manage them according to their hazardous-waste classification. • Inspect sumps, drains, interceptors and wastewater-treatment systems for Pb-		

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		bearing sludge and acidic residues. Collect contaminated liquids and sludges for appropriate treatment/ disposal. Prevent contaminated decommissioning water from entering stormwater systems. • Develop a waste inventory and classify wastes before removal. Likely waste streams include Pb-bearing dust, sludge, contaminated PPE, spent filters, contaminated soil, electrolyte, chemicals, batteries, contaminated scrap and wastewater-treatment residues. Store wastes in labelled, compatible, sealed/ covered containers with secondary containment. Maintain waste manifests and disposal/ recycling records. • Segregate clean scrap from Pb-contaminated scrap. Do not release contaminated metal as ordinary scrap until it has been appropriately characterised and, where required, decontaminated. Use authorised recyclers for Pb-containing materials. • Investigate areas around battery storage, manufacturing lines, paste preparation, casting, charging, waste storage, loading areas, drainage channels and historical spill locations. If Pb contamination is identified, delineate the contamination spatially and vertically. Depending on risk and applicable regulatory criteria, options can include excavation and off-site treatment/ disposal, stabilisation/ solidification, containment, or other approved remediation technologies. The Environmental Protection Agency (United States) experience with lead-acid battery sites demonstrates the importance of assessing and remediating contaminated soil and groundwater. • Investigate potential groundwater contamination where historical releases or permeable soils indicate a pathway. Prevent further infiltration from contaminated areas and establish groundwater monitoring where warranted by the site assessment. • Separate clean and potentially contaminated stormwater during decommissioning. Cover stockpiles and waste-storage areas, maintain bunds and drainage controls, and inspect drainage after rainfall. Prevent sediment and Pb-contaminated runoff from leaving the site. • Establish a Pb/ dust monitoring programme appropriate to the level of contamination and work activities. Monitor at work areas and, where necessary, site boundaries. Stop or modify activities if monitoring indicates unacceptable releases. • Maintain spill kits suitable for acid and contaminated liquids. Establish procedures for acid spills, Pb-containing dust releases, contaminated-water releases and damaged batteries. Ensure emergency response arrangements remain operational throughout decommissioning. • Package and label hazardous materials appropriately and use licensed/ authorised transporters where required. Cover loads and prevent leakage or dust emissions during transport. Maintain records showing destination and final treatment/recycling/ disposal. • Sequence demolition so that hazardous materials and contaminated equipment are		
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	ASPECT	RECOMMENDED MITIGATION MEASURES	RESPONSIBLE PARTY	FREQUENCY OF ACTION
		removed before general demolition. Isolate contaminated areas, suppress dust and prevent contaminated demolition rubble from being mixed with “cleaner” waste. • Treat disposable PPE, filters, wipes, absorbents and cleaning materials from contaminated areas as potentially Pb-contaminated until characterised. Prevent workers from taking contaminated clothing or equipment off-site. WHO specifically highlights the risk of workers transferring Pb contamination from workplaces to homes. • After dismantling, conduct systematic cleaning using wet methods and/or HEPA vacuuming. Remove residual Pb-bearing deposits from floors, drains, structural surfaces and equipment. Undertake confirmatory environmental sampling to demonstrate that applicable cleanup criteria have been achieved. • Prepare a final environmental closure report documenting decommissioning activities, quantities and destinations of hazardous materials/ wastes, analytical results, remediation undertaken, incidents, monitoring data and confirmation that environmental closure requirements have been met. The International Finance Corporation’s General Environmental, Health, and Safety Guidelines specifically identify contaminated land, hazardous-material management, waste management, wastewater, air emissions and decommissioning as relevant environmental-management areas. • Secure the site, control public access, and ensure that Pb-containing materials, spent batteries, contaminated soil, and hazardous wastes are properly contained, transported, and treated/ recycled by authorised facilities. • Develop a traffic-management plan for trucks transporting equipment, scrap metal, batteries, and hazardous waste. Schedule heavy vehicle movements to minimise disruption to surrounding communities. • Ensure prompt site clean-up, remediation, rehabilitation, and appropriate redevelopment/ reuse so that the closed facility does not become a source of blight or perceived contamination. • Conduct contamination assessments and remediate areas affected by Pb, acid, hydrocarbons or other hazardous substances. Maintain appropriate post-closure monitoring where contamination remains. • Retain suitably qualified personnel during critical decommissioning stages to ensure safe shutdown, hazardous-material identification, equipment isolation, and knowledge transfer. • Monitor relevant parameters throughout decommissioning, potentially including airborne particulate/ Pb, soil Pb, surface water, groundwater, wastewater pH and metals, and waste characteristics. Monitoring should be risk-based and linked to the identified contamination pathways.		

6. ROLES AND RESPONSIBILITIES

6.1 PROJECT COORDINATOR

The Project Coordinator is responsible for overall management of the project and the implementation of the EMPr. The following tasks fall within his/ her responsibilities:

- Be familiar with the recommendations and mitigation measures of this EMPr, and implement these measures;
- Monitor site activities on a daily basis for compliance;
- Conduct internal audits of the construction site against the EMPr;
- Confine the construction site to the demarcated areas; and
- Rectify transgressions through the implementation of corrective action.

6.2 ENVIRONMENTAL CONTROL OFFICER

The Developer must appoint a suitable qualified Environmental Officer that will be responsible for ensuring compliance to the EMPr. The ECO on site will conduct regular site visits to ensure the success of the EMPr. The ECO will:

- Do regular site visits to be able to compile site inspection reports for the inclusion in the EMPr as an addendum if necessary.
- Report compliance and non-compliance to the CA.
- Know the contents and implications of the environmental report and monitor the implementation of the findings using the EMPr.
- Act as a guide, advisor and consultant to the contractor and client on environment issues during construction. This will be achieved by continuous auditing of the project, identification of problem areas and provisioning of action plans to avoid costly stoppages and/ environmental damage.
- Ensure that people can report incidents and recommend corrective action when required for aspects of non-compliance within the EMPr.
- Upgrade the EMPr as necessary and inform the relevant parties of the changes.

6.3 CONTRACTOR

The construction contractor is responsible for submitting reports on implementation of the environmental plan as described by the project proponent and authorities. The contractor must always ensure compliance with the EMPr during construction activities and maintain an environmental register which keeps a record of all environmental incidents that occur such as:

- Public involvement/ complaints;
- Health and safety incidents;
- Incidents involving hazardous materials stored on site; and
- Non-compliance incidents.

7. IMPLEMENTATION OF ENVIRONMENTAL MANAGEMENT PROGRAMME

A copy of the EMPr must be kept on site at all times during the construction period. The EMPr will be binding on all contractors operating on the site and must be included within the Contractual Clauses. The monitoring programme must be implemented for the duration of the construction and operation of the sport facility. This programme should include:

- Establishing a baseline of pre-construction site conditions validated with photographic evidence.

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- Monthly audits will be conducted by an independent ECO for the construction phase to ensure compliance with the conditions stipulated in this EMPr and, where necessary, make recommendations for corrective action. These audits can be conducted randomly and do not require prior arrangement with the Project Coordinator.
- Compilation of an audit report with a rating of compliance with the EMPr. The ECO must keep a photographic record of the demarcated site and construction area. The Contractor must be held liable for all unnecessary damage to the environment. A register must be kept of all complaints from the community. All complaints must be handled immediately to ensure timeous rectification by the responsible party.

7.1 ENVIRONMENTAL AUDITS AND MONITORING

The contractor will submit reports on implementation of the environmental plan as described by the project proponent and authorities. An environmental audit will be conducted prior to the development and construction activities. This environmental audit will ensure that:

- Mitigation measures are implemented as prescribed in the EMPr;
- The relevant authorities are kept informed about progress with the project and that they are given assurance that the project is implemented and managed as prescribed in the EMPr;
- Periodic inspections and/or audits are performed;
- Compliance to the conditions of the EMPr is adhered to and a report compiled;
- Reviews of conformance against policies and procedures stated in this document are performed;
- Supervisors in all work areas will conduct performance and compliance reviews, using the EMPr as guideline to ensure compliance; and
- Inspections will occur monthly (or as required).

7.2 RECORD KEEPING

Documents to be maintained by the designated representative/ site agent are to include:

- Training records;
- Inspection records;
- Records of non-compliance and corrective action;
- Records of all complaints, concerns or issues and corrective action;
- EMPr and
- All incidents' reports.

All records will be kept for up to a year after the completion of the project or in accordance with other legal requirements as they apply.

7.3 EMPr UPDATES

The EMPr will be subject to on-going review throughout the course of the project to ensure its continued suitability, adequacy, and effectiveness. This review may include but will not be limited to monitoring and measuring information, performance data, assessment and audit results and other relevant information and data.

7.4 NON-COMPLIANCE

The contractors must act immediately when notice of non-compliance is received and take corrective action. Complaints received regarding activities on the construction site pertaining to the environment must be recorded in a dedicated register and the response(s) noted with the date and action taken. The ECO should be made aware of any complaints.

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Any non-compliance with the agreed procedures of the EMPr is a transgression of the various statutes and laws that define how the environment is managed. Failure to redress the cause must be reported to the CA for them to deal with the transgression, as it deems fit. The contractor/ sub-contractor is deemed not to have complied with the EMPr if, inter alia:

- There is evidence of contravention of the EMPr specifications within the boundaries of the construction site and site extensions;
- There is contravention of the EMPr specifications which relate to activities outside the boundaries of the construction site;
- Environmental damage ensues due to negligence;
- Construction activities take place outside the defined boundaries of the site; and/
- The contractor/ sub-contractor fails to comply with corrective or other instructions issued within a specific time.

It is recommended that the Contractors institute penalties for the following less serious violations and any others determined during work, as detailed below:

- Littering on site.
- Lighting of illegal fires on site.
- Persistent or unrepaired fuel and oil leaks.
- Any persons, vehicles or equipment related to the contractor's/ sub-contractor's operations found within the designated "no-go" areas.
- Excess dust or excess noise emanating from site.
- Possession or use of intoxicating substances on site.
- Any vehicles being driven more than designated speed limits.
- Removal and/or damage to fauna, flora, cultural or heritage objects on site.
- Urination and defecation anywhere except at designated facilities.

7.5 INCIDENT REPORTING AND REMEDY

If a leakage or spillage of hazardous substances occurs on site, the local emergency services must be immediately notified of the incident. The following information must be provided:

- the location;
- the nature of the load;
- the extent of the impact; and
- the status at the site of the accident itself (i.e., whether further leakage is still taking place, whether the vehicle or the load is on fire).

Written records must be kept on the corrective and remedial measures decided upon and the progress achieved therewith over time. Such progress reporting is important for monitoring and auditing purposes. The written reports may be used for training purposes to prevent similar future occurrences.

7.6 PENALTIES

Where environmental damage is caused or a pollution incident, and/or failure to comply with any of the environmental specifications contained in the EMPr, the project proponent/ site manager and/ the contractor/ sub-contractor will be liable. The following violations, and any others determined during work, should be penalised:

- Hazardous chemical/ oil spill and/ dumping in non-approved sites.

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- Damage to sensitive environments.
- Damage to cultural and historical sites
- Unauthorised removal/ damage to indigenous trees and other vegetation, particularly in identified sensitive areas.
- Uncontrolled/ unmanaged erosion.
- Unauthorised blasting activities (if applicable).
- Pollution of water sources.
- Unnecessary removal or damage to trees.

8. SUMMARY OF RECOMMENDATIONS AND CONCLUSION

The proposed development will have minimal environmental impacts which are manageable through good engineering practices and following all environmental recommendations prescribed. Although all foreseeable actions and potential mitigations or management actions are contained in this document, the EMPr should be considered as a day-to-day management document. The EMPr thus sets out the environmental standards that are required to minimise the negative impacts and maximise the positive benefits of the local communities. An EMPr is a “live document”, and its continuous review and correct management will definitely result to the successful construction of the proposed development.

The contractor is to be made aware of the potential cost and timing implications needed to fulfil the implementation of the EMPr, thus adequately costing for these.

ANNEXURE 1: REHABILITATION SCHEDULE AND GUIDELINE

Where applicable, the rehabilitation schedule and procedure to be adhered to is as follows (this methodology should be updated during the construction phase):

Step	Method	Equipment
1	Remove all construction material from the area where construction has been completed	To be undertaken by hand.
2	The ground should be sloped so as to attain a natural slope and to attain a natural water flow, if it has been altered during construction (the natural slope should be altered as little as possible during construction).	To be undertaken by hand.
3	Topsoil that has been stockpiled during construction must be applied to the area to undergo rehabilitation. The depth of the topsoil layer to be applied depends on the natural depth of topsoil in the area, and the amount of topsoil that may have been lost during construction.	Topsoil must be applied from the topsoil stockpiled during construction
4	The area should be mulched to improve water retention, and brushwood applied to act as a soil stabiliser. Mulch and brushwood must be applied more heavily in areas which are presently well-wooded (or as directed by the ECO).	The mulch used should be woodchip, obtained commercially or from trees removed during site clearance. The brushwood is obtained from the bushes and trees removed during site clearance.
5	The planting of locally indigenous species should be undertaken, with local species endemic to the area.	The plant species to be used during the rehabilitation process should be endemic to the area surrounding the substation installation.
6	The species which has been planted should be regularly watered after planting. Watering should ensure that no erosion of the topsoil takes place.	A hosepipe must be available on site.
7	Plant species which are not well established after a period of 2 months after planting should be re-planted. If necessary, another dressing of topsoil should be applied prior to planting.	As above
8	Slope stabilisation measures may be necessary in areas where vegetation has not been able to establish and there is an erosion risk. The measures implemented depend on the situation, and can be varied as necessary. The ECO will direct the Contractor as required.	Various slope stabilisation measures are available and vary in effectiveness according to the situation including • Onion bags • Soil saver (jute/hessian) blankets • Logs/bark held in place with pegs

ANNEXURE 2: PROPOSED FINES AND PENALTY STRUCTURE

Typical incidents incurring penalties	Value
Discharging effluent onto the ground or into stormwater channels (per incident)	R2,000
Failure to provide adequate sanitation, waste disposal facilities or services (per incident)	R1,000
No induction regarding environmental matters and site housekeeping practices (per employee)	R2,000
Failure to provide adequate equipment for emergency situations (per incident)	R5,000
Failure to maintain basic safety measures on site (per incident)	R1,000
Inappropriate use of bins and poor waste management on site (per incident)	R1,000
Inappropriate off-site disposal of waste from site (per incident)	R10,000
Failure to maintain and register incidents in the Incident Register (per incident)	R1,000
Any contravention with a Method Statement (per incident)	R5,000
Repeated contravention of the specifications or failure to comply with instructions (per incident)	R5,000

ANNEXURE 3: ISSUES TO BE ADDRESSED BY AND WHERE APPROPRIATE SHOWN ON THE ENVIRONMENTAL MANAGEMENT SITE PLAN

Issue	Nature / Description
Sequence of events	Description of the nature of the process required. Briefly describe the sequence of events that will take place from the time that the contractor moves onto site to the time when the site is handed over to the Project Proponent
Health and safety	Potential risks and hazards and precautions that will be taken Cooking area, hazardous materials site, first aid kit, fuel store, security issues, fire management, and public safety.
On site toilets	How many required for the particular development? Are there existing ablution facilities that can be used? If not, how many toilets need to be brought to site and how long are the toilets required on site? Location of toilets (Site Plan)
Workforce	Number of onsite workers Training of workforce in terms of environmental awareness Management of workforce, particularly sub-contractors
Transport and traffic	Transport required for site workers Routes to be used by construction vehicles Demarcate location of traffic turning circle and parking areas (Site Plan)
Infrastructure and associated equipment	Nature and extent of infrastructure needed for construction Indicate on site plan
Material storage (e.g. sand needed to build wall)	Approximation of quantity to be excavated Where to be stockpiled (Site Plan) How long to be stockpiled Area required for stockpile
Earthworks/cleaning	Volume of material to be excavated/cleaned Duration of operations Where stocks to be kept on site (Site Plan) How long to be kept on site Where, when and how to be disposed of Recycling and/or re-use of materials
Equipment needed for construction activities	Area required for material and equipment storage Duration of works Nature of equipment and necessary materials
Drinking water	Quantity required Duration of period in which required Source of water Location of potable water (Site Plan)
Cooking and rest areas	Area required Equipment required e.g. gas stoves, matches etc. Shelter Location – indicate on site plan
Existing structures	Indication of location of any structures that need to be removed and/or protected
Life of project	Working hours Time frame Phasing of work areas
Construction site	Work area required Location of construction site and work area (Site Plan) Demarcation of no-go areas
Environmentally sensitive areas and possible environmental risks associated with construction activities	A plan of environmentally sensitive areas must be drawn up and made available to contractors. This will include the riverine corridor and pan demarcated in the field survey. A training programme on possible environmental risks that may result from construction activities and how to deal with these (including a reporting structure) must be made available prior to construction commencing.

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Issue	Nature / Description
Waste management	Litter drums - number, type, size, location (Site Plan) Closest registered waste disposal site (Location map) Waste management plan Recycling/ material re-use options

ANNEXURE 4: GENERAL CODE OF CONDUCT FOR CONTRACTORS

Contractors shall ensure that all sub-contractors, employees, suppliers, agents, etc. are fully aware of the environmental issues detailed in the EMPr.

Contractors must Investigate and comply with all existing regulations and laws/ bylaws unless the relevant authority grants specific written authority waiving compliance with any legislation.

Contractor's, *inter alia*, commit to the following code of conduct:

- Implementing, documenting, reviewing, and complying with environmental specifications, permits, and management plans as well as relevant environmental legislation, regulations, and standards.
 - Ensuring the communication and training of all personnel through induction and appropriate environmental training.
 - Reducing and, where possible, eliminating and preventing any harmful effects to the environment and societal milieu.
 - Liaising openly with, and reporting publicly to, all interested and affected parties on environmental matters.
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ANNEXURE 5: ENVIRONMENTAL INCIDENT REGISTER

ENVIRONMENTAL INCIDENT REGISTER				
Date	Environmental incident	Comments <i>(Include any possible explanations for current condition and possible responsible parties. Include photographs, records etc. if available)</i>	Corrective Action Taken <i>(Give details and attach documentation as far as possible)</i>	Signature

ANNEXURE 6: ENVIRONMENTAL COMPLAINTS REGISTER

ENVIRONMENTAL COMPLAINTS REGISTER				
Date	Environmental Complaint	Name of Complainer (Name of complained and method which complaint was received)	Corrective Action Taken (Give details and attach documentation as far as possible)	Signature

REFERENCES

- BGIS. Biodiversity spatial datasets. Available at: <https://bgis.sanbi.org>
- Department of Forestry, Fisheries and the Environment, 2020. General requirements for undertaking site sensitivity verification for environmental themes. Government Gazette No. 43110. Available at: <https://www.gov.za/documents/national-environmental-management-act-procedures-assessment-and-minimum-criteria-0>
- Department of Forestry, Fisheries and the Environment, 2020. Protocol for Specialist Assessment and Minimum Report Content Requirements. Available at: <https://www.gov.za/documents/national-environmental-management-act-procedures-assessment-and-minimum-criteria-0>
- Eastern Cape Biodiversity Conservation Plan, 2019. Eastern Cape Biodiversity Conservation Plan. Available at: <https://ecbcplan.co.za>
- Eastern Cape Parks and Tourism Agency 'Eastern Cape Biodiversity Conservation Plan' 2019.
- Grobler, D., 2012. Climatic conditions of the Nelson Mandela Bay area.
- McCallum, D., 1981. Climatic characteristics of the Port Elizabeth region.
- Muir et al (2017) 'Lithostratigraphy of the Kirkwood Formation (Uitenhage Group), including the Bethelsdorp, Colchester and Swartkops Members, South Africa'.
- Schumann, E.H. et al., 1991. Wind patterns in the Port Elizabeth region.
- Schumann, E.H. et al., 1999. Seasonal wind variability in the Eastern Cape coastal region. *Journal of Marine Research* 57, (4). https://elischolar.library.yale.edu/journal_of_marine_research/2332
- South African National Biodiversity Institute, 2024. Vegetation Map of South Africa. Available at: <https://bgis.sanbi.org>
- Stone, A., 1988. Climate of the Eastern Cape coastal region.
- South African National Biodiversity Institute) (2006) 'National Spatial Biodiversity Assessment 2004: Vol. 1: Terrestrial Component'.
- Vogelgat Private Nature Reserve (n.d.) 'Geology & Geomorphology'.